

Unit 2. Elementary Matrices and Vector Spaces

1. Row operations as matrix multiplication

For $1 \leq r, s \leq m$, let M_{rs} be the $m \times m$ matrix with a single 1 in position (r, s) and zeros elsewhere. Write A_i for row i of an $m \times n$ matrix A . By the definition of matrix multiplication,

$$(M_{rs}A)_{ij} = \sum_{k=1}^m (M_{rs})_{ik} a_{kj} = \begin{cases} a_{sj}, & i = r, \\ 0, & i \neq r. \end{cases}$$

Every row of M_{rs} except row r is zero; its row r selects the s th row of A . Thus $M_{rs}A$ puts A_s in position r and puts zero in every other row. In particular, $M_{rr}A$ keeps row r and replaces all the other rows by zero.

Example 2.1 (Selecting and interchanging rows). For

$$A = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 4 & 2 \\ -2 & 5 & 1 \end{pmatrix},$$

row selection gives

$$M_{12}A = \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_{21}A = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Their sum swaps the first two rows but deletes the third:

$$(M_{12} + M_{21})A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} A = \begin{pmatrix} 1 & 4 & 2 \\ 3 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Adding $M_{33}A$ restores the third row. Therefore

$$(M_{12} + M_{21} + M_{33})A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & 4 & 2 \\ 3 & 2 & -1 \\ -2 & 5 & 1 \end{pmatrix}.$$

The left factor is obtained by swapping rows 1, 2 of I_3 .

Proposition 2.1 (The three elementary matrices). Applying one elementary row operation to I_m produces a matrix E such that EA performs that same operation on every $m \times n$ matrix A . Specifically,

$$\begin{aligned} S_{rs} &= M_{rs} + M_{sr} + \sum_{i \notin \{r,s\}} M_{ii} \quad (r \neq s), \quad R_r \leftrightarrow R_s, \\ D_r(c) &= I_m + (c-1)M_{rr} \quad (c \neq 0), \quad R_r \leftarrow cR_r, \\ T_{rs}(c) &= I_m + cM_{rs} \quad (r \neq s), \quad R_r \leftarrow R_r + cR_s. \end{aligned}$$

Proof. For the first matrix, distribute the product:

$$S_{rs}A = M_{rs}A + M_{sr}A + \sum_{i \notin \{r,s\}} M_{ii}A.$$

The first two summands put A_s in row r and A_r in row s ; each remaining summand restores one of the unchanged rows.

For scaling, $D_r(c)A = A + (c-1)M_{rr}A$. Its row r is $A_r + (c-1)A_r = cA_r$, and every other row is A_i . For addition, $T_{rs}(c)A = A + cM_{rs}A$. Its row r is $A_r + cA_s$, and every other row is A_i . $\square$

The non-diagonal entry c of $T_{rs}(c)$ is in position (r, s) : r is the row being changed and s supplies the row being added. For example,

$$T_{13}(4) = \begin{pmatrix} 1 & 0 & 4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

adds four times row 3 to row 1 of a matrix with four rows.

Proposition 2.2 (Elementary inverses).

$$S_{rs}^{-1} = S_{rs}, \quad D_r(c)^{-1} = D_r(c^{-1}), \quad T_{rs}(c)^{-1} = T_{rs}(-c).$$

Proof. Swapping the same two rows twice restores every row. Applying this to I_m gives $S_{rs}S_{rs} = I_m$. Multiplying row r by c and then by c^{-1} restores that row; the reverse order does too. Thus

$$D_r(c^{-1})D_r(c) = D_r(c)D_r(c^{-1}) = I_m.$$

The addition $R_r \leftarrow R_r + cR_s$ leaves row s unchanged because $r \neq s$. It is undone by $R_r \leftarrow R_r - cR_s$:

$$(A_r + cA_s) - cA_s = A_r.$$

Starting with the subtraction gives $(A_r - cA_s) + cA_s = A_r$. Applying both orders to I_m proves $T_{rs}(-c)T_{rs}(c) = T_{rs}(c)T_{rs}(-c) = I_m$. $\square$

The row-selection calculation also determines products of matrix units:

$$M_{rs}M_{uv} = \begin{cases} M_{rv}, & s = u, \\ 0, & s \neq u. \end{cases}$$

Indeed, row s of M_{uv} is zero unless $s = u$; in that case it has its 1 in column v . In particular, $M_{rr}^2 = M_{rr}$ and $M_{rs}^2 = 0$ for $r \neq s$. Hence

$$T_{rs}(c)T_{rs}(d) = (I + cM_{rs})(I + dM_{rs}) = I + (c + d)M_{rs} = T_{rs}(c + d).$$

If e_r denotes a standard coordinate column, entrywise multiplication gives $M_{rs} = e_re_s^\top$. Thus the outer-product notation describes the same row selectors already computed above.

Example 2.2 (Operation order). For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$, the successive operations $R_1 \leftrightarrow R_3$, $R_2 \leftarrow 7R_2$, and $R_3 \leftarrow R_3 + 5R_2$ give

$$A \mapsto \begin{pmatrix} 5 & 6 \\ 3 & 4 \\ 1 & 2 \end{pmatrix} \mapsto \begin{pmatrix} 5 & 6 \\ 21 & 28 \\ 1 & 2 \end{pmatrix} \mapsto \begin{pmatrix} 5 & 6 \\ 21 & 28 \\ 106 & 142 \end{pmatrix}.$$

The final matrix is $T_{32}(5)D_2(7)S_{13}A$. The first operation is the factor closest to A . To undo the calculation, multiply on the left by $S_{13}D_2(1/7)T_{32}(-5)$.

Column operations are right multiplication. Here use $n \times n$ matrix units, since A has n columns. The entry calculation is

$$(AM_{rs})_{ij} = \sum_{k=1}^n a_{ik}(M_{rs})_{kj} = \begin{cases} a_{ir}, & j = s, \\ 0, & j \neq s. \end{cases}$$

Thus AM_{rs} puts column r of A in column s and zero elsewhere. Consequently $A(I_n + cM_{rs})$ adds c times column r to column s . The same entry (r, s) changes row r on the left and column s on the right. If F is invertible, the correspondence $x = Fz$, $z = F^{-1}x$ gives

$$AFz = b \iff Ax = b.$$

The coordinates of the unknown have changed; the original solution columns x need not also solve $AFx = b$.

2. Invertibility and the inverse algorithm

Proposition 2.3 (Inverses of products). If $A_1, \dots, A_k$ are invertible square matrices of the same size, then

$$(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}.$$

Proof. For two factors, associativity permits the adjacent cancellations

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I.$$

For k factors, in $(A_1 \cdots A_k)(A_k^{-1} \cdots A_1^{-1})$ first cancel $A_k A_k^{-1}$, then $A_{k-1} A_{k-1}^{-1}$, and continue to $A_1 A_1^{-1}$. The reversed product cancels starting with $A_1^{-1} A_1$. Both products are I , as required. $\square$

If successive row operations multiply by $E_1, \dots, E_k$, then

$$A \mapsto E_1 A \mapsto E_2 E_1 A \mapsto \cdots \mapsto H = E_k \cdots E_1 A = EA.$$

Every E_i is invertible, so $E^{-1} = E_1^{-1} \cdots E_k^{-1}$. Multiplying by E and E^{-1} gives

$$Ax = b \iff Hx = Eb.$$

For square A , if A^{-1} exists, then $(EA)^{-1} = A^{-1}E^{-1}$. Conversely, if H^{-1} exists, then $A = E^{-1}H$ is a product of invertible matrices and $A^{-1} = H^{-1}E$. Thus row equivalent square matrices are either both invertible or both noninvertible.

Proposition 2.4 (From an upper triangular matrix to the identity). Every upper triangular square matrix with nonzero diagonal entries is row equivalent to the identity.

Proof. Write the matrix as

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{pmatrix}, \quad u_{ii} \neq 0.$$

Perform $R_i \leftarrow u_{ii}^{-1} R_i$ for every i . With $t_{ij} = u_{ij}/u_{ii}$ this gives

$$T = \begin{pmatrix} 1 & t_{12} & \cdots & t_{1,n-1} & t_{1n} \\ 0 & 1 & \cdots & t_{2,n-1} & t_{2n} \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & t_{n-1,n} \\ 0 & \cdots & 0 & 0 & 1 \end{pmatrix}.$$

Its last row is $e_n^\top$. For $i = 1, \dots, n-1$, perform $R_i \leftarrow R_i - t_{in} R_n$. Only entry (i, n) changes, becoming $t_{in} - t_{in} = 0$. The result has the form

$$\begin{pmatrix} 1 & t_{12} & \cdots & t_{1,n-1} & 0 \\ 0 & 1 & \cdots & t_{2,n-1} & 0 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & 0 \\ 0 & \cdots & 0 & 0 & 1 \end{pmatrix}.$$

Now row $n-1$ is $e_{n-1}^\top$. Subtract $t_{i,n-1}$ times that row from row i , for $i < n-1$. This clears column $n-1$ above its diagonal without changing any other column. More generally, after columns $j+1, \dots, n$ have been cleared, row j is $e_j^\top$: its earlier entries are zero by triangularity and its later entries have just been cleared. Subtract the current entry (i, j) times row j from row i , for each $i < j$. Proceeding with $j = n, n-1, \dots, 2$ therefore gives I_n . For $n = 1$, the initial scaling already gives I_1 . $\square$

For order three the entire procedure reads

$$\begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix} \mapsto \begin{pmatrix} 1 & b/a & c/a \\ 0 & 1 & e/d \\ 0 & 0 & 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 & b/a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mapsto I_3,$$

where $a, d, f \neq 0$. After scaling, the operations are $R_1 \leftarrow R_1 - (c/a)R_3$, $R_2 \leftarrow R_2 - (e/d)R_3$, and $R_1 \leftarrow R_1 - (b/a)R_2$.

Theorem 2.5 (Square inverse criterion). For an $n \times n$ matrix A , the following statements are equivalent:

- (a) A has a two-sided inverse.
- (b) A is row equivalent to I_n .
- (c) $Ax = 0$ has only the solution $x = 0$.
- (d) For every $b \in \mathbb{R}^n$, $Ax = b$ has exactly one solution.

When these hold, A is a product of elementary matrices and $x = A^{-1}b$.

Proof. Reduce A to an echelon matrix $H = EA$, using the elimination theorem from Unit 1. If H has a zero row, every product HC has that zero row, so H cannot have a right inverse. Thus (a) forces n nonzero rows. Their strictly increasing pivot columns are n distinct indices in $\{1, \dots, n\}$, so they must be $1, \dots, n$ in that order. Therefore H is upper triangular with nonzero diagonal. Proposition 2.4 reduces it to I_n , proving (b).

If $FA = I_n$ is the accumulated row reduction, F is a product of elementary matrices and hence invertible. Multiplication by F^{-1} gives $A = F^{-1}$, so $F = A^{-1}$ and (b) implies (a). Moreover, F^{-1} is a product of the elementary inverses in reverse order.

From (a), $Ax = 0$ implies $x = A^{-1}0 = 0$. Conversely, if an echelon form of A has fewer than n pivots, a free variable can be set to 1, the other free variables to 0, and back substitution produces a nonzero solution of $Ax = 0$. Thus (c) forces n pivots and implies (b). Finally, if (a) holds, $A(A^{-1}b) = (AA^{-1})b = b$, so $A^{-1}b$ is a solution. If $Ax = b$, then

$$x = (A^{-1}A)x = A^{-1}(Ax) = A^{-1}b,$$

which proves uniqueness and (d). Statement (d), applied to $b = 0$, implies (c). □

To compute A^{-1} , reduce the *left block* of $[A \mid I_n]$ to I_n . Each operation is applied across the entire augmented matrix:

$$[A \mid I_n] \mapsto [EA \mid E] = [I_n \mid A^{-1}].$$

If the left block has fewer than n pivots, A is not invertible. Pivots created in the right block cannot repair this failure.

Example 2.3 (A complete inverse calculation). Let $A = \begin{pmatrix} 2 & -3 & 1 \\ 1 & 1 & -1 \\ 2 & 0 & 1 \end{pmatrix}$. Start with $[A \mid I_3]$ and perform $R_1 \leftrightarrow R_2$:

$$\left[\begin{array}{ccc|ccc} 2 & -3 & 1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \mapsto \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 2 & -3 & 1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

Next perform $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 2R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -5 & 3 & 1 & -2 & 0 \\ 0 & -2 & 3 & 0 & -2 & 1 \end{array} \right].$$

The operation $R_3 \leftarrow R_3 - \frac{2}{5}R_2$ gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -5 & 3 & 1 & -2 & 0 \\ 0 & 0 & 9/5 & -2/5 & -6/5 & 1 \end{array} \right].$$

The last pivot is $9/5$. To clear the entries above it, use $R_2 \leftarrow R_2 - \frac{5}{3}R_3$ and $R_1 \leftarrow R_1 + \frac{5}{9}R_3$. This gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -2/9 & 1/3 & 5/9 \\ 0 & -5 & 0 & 5/3 & 0 & -5/3 \\ 0 & 0 & 9/5 & -2/5 & -6/5 & 1 \end{array} \right].$$

Now $R_1 \leftarrow R_1 + \frac{1}{5}R_2$ clears the entry above the second pivot:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/9 & 1/3 & 2/9 \\ 0 & -5 & 0 & 5/3 & 0 & -5/3 \\ 0 & 0 & 9/5 & -2/5 & -6/5 & 1 \end{array} \right].$$

Finally perform $R_2 \leftarrow -\frac{1}{5}R_2$ and $R_3 \leftarrow \frac{5}{9}R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/9 & 1/3 & 2/9 \\ 0 & 1 & 0 & -1/3 & 0 & 1/3 \\ 0 & 0 & 1 & -2/9 & -2/3 & 5/9 \end{array} \right].$$

Thus the right block is A^{-1} . Its three columns satisfy

$$A \frac{1}{9} \begin{pmatrix} 1 \\ -3 \\ -2 \end{pmatrix} = e_1, \quad A \frac{1}{3} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = e_2, \quad A \frac{1}{9} \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} = e_3.$$

For an arbitrary right-hand side, multiplication by the inverse gives

$$x = A^{-1}b = \frac{1}{9} \begin{pmatrix} b_1 + 3b_2 + 2b_3 \\ -3b_1 + 3b_3 \\ -2b_1 - 6b_2 + 5b_3 \end{pmatrix}.$$

Further inverse results: reading

Proposition 2.6 (A one-sided inverse for a square matrix is two-sided). If A, B are $n \times n$ and either $BA = I_n$ or $AB = I_n$, then $B = A^{-1}$ and both identities hold.

Proof. If $BA = I_n$, then $Ax = 0$ gives $x = BAx = 0$, so the theorem gives A^{-1} . Now $B = BAA^{-1} = A^{-1}$. If $AB = I_n$, apply the result just proved to $B^T A^T = I_n$, and transpose the resulting identity $A^T B^T = I_n$. $\square$

The square hypothesis matters. For $A = (1 \ 0)$ and $B = (1 \ 0)^T$, $AB = 1$ but $BA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2$.

Example 2.4 (The inverse fails at an exceptional parameter). For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ put $\Delta = ad - bc$. If $\Delta \neq 0$, direct multiplication in both orders gives

$$A^{-1} = \frac{1}{\Delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $\Delta = 0$ and the first row is nonzero, $(b, -a)^T \neq 0$ is killed by A . If the first row is zero and the second is nonzero, use $(d, -c)^T$; if $A = 0$, use e_1 . Thus A is invertible exactly when $\Delta \neq 0$. This is a direct two-variable calculation; no general determinant theorem is needed.

3. Columns, relations, and generated sets

For columns $a_1, \dots, a_n \in \mathbb{R}^m$, an expression $x_1 a_1 + \dots + x_n a_n$ is a **linear combination**, with **coefficients** $x_1, \dots, x_n$. It is **nontrivial** if at least one coefficient is nonzero, even when the resulting vector is zero. Write $A = (a_1 \cdots a_n)$. Expanding the matrix product gives

$$Ax = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{pmatrix} = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} = \sum_{j=1}^n x_j a_j.$$

Thus $Ax = b$ asks whether b is a linear combination of the columns. A **relation** among the columns is an equality $\sum_j x_j a_j = 0$. The homogeneous system has three equivalent readings:

$$\begin{aligned} Ax = 0 &\iff \sum_{j=1}^n x_j a_j = 0 && \text{a relation among columns,} \\ &\iff A_i \cdot x = 0 \quad (i = 1, \dots, m) && \text{perpendicularity to every row.} \end{aligned}$$

Here the row $A_i = (a_{i1}, \dots, a_{in})$ is viewed as a vector of $\mathbb{R}^n$ in the dot product. The last equivalence holds because $(Ax)_i = \sum_j a_{ij}x_j = A_i \cdot x$.

The columns are **linearly dependent** if they have a nontrivial relation, and **linearly independent** otherwise. In symbols, independence means

$$x_1 a_1 + \cdots + x_n a_n = 0 \implies x_1 = \cdots = x_n = 0.$$

For square A , this is equivalent to invertibility by Theorem 2.5. The standard vectors are independent:

$$x_1 e_1 + \cdots + x_n e_n = (x_1, \dots, x_n)^\top = 0 \implies x_i = 0 \text{ for every } i.$$

The same calculation identifies every column of a product. If $B = (b_{jk})$ has n rows and $C = AB$, then column k is

$$c_k = A \begin{pmatrix} b_{1k} \\ \vdots \\ b_{nk} \end{pmatrix} = b_{1k} a_1 + \cdots + b_{nk} a_n.$$

Indeed the i th coordinate on either side is $\sum_{j=1}^n a_{ij} b_{jk} = c_{ik}$.

If $H = EA$ is a row reduction, its columns are $h_j = E a_j$. For every choice of coefficients,

$$\sum_j x_j h_j = \sum_j x_j E a_j = E \left(\sum_j x_j a_j \right).$$

Thus a relation among the original columns gives a relation with the same coefficients among the new columns. Conversely,

$$\sum_j x_j h_j = 0 \implies \sum_j x_j a_j = E^{-1} \left(\sum_j x_j h_j \right) = 0.$$

Row reduction preserves every column relation, coefficient by coefficient. It need not preserve the set of column combinations itself: that set is multiplied by E .

Example 2.5 (Worked reading: redundant generators). Let

$$A = \begin{pmatrix} 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -2 & 0 & 4 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix}.$$

Swap rows 1, 2, then perform $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 - R_2$, using the current rows:

$$A \mapsto \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix} \mapsto \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix} \mapsto \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Halving row 1 gives

$$H = \begin{pmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Its columns satisfy $h_2 = 2h_1$, $h_4 = 2h_1 + 2h_3$, and $h_5 = 5h_1 + 3h_3$. Hence the original columns satisfy

$$a_2 = 2a_1, \quad a_4 = 2a_1 + 2a_3, \quad a_5 = 5a_1 + 3a_3.$$

All five columns therefore generate the same set as a_1, a_3 . The equation $sa_1 + ta_3 = (u, v, w)^\top$ is

$$t = u, \quad 2s - 2t = v, \quad 2s - t = w.$$

The first two equations give $t = u$ and $s = v/2 + u$; the third then requires $w = u + v$. Conversely, if $w = u + v$, those choices of s, t satisfy all three equations. The generated set is exactly the plane $w = u + v$. To find all relations, the nonzero rows of $Hx = 0$ give

$$x_1 + 2x_2 - x_3 + 2x_5 = 0, \quad x_3 + 2x_4 + 3x_5 = 0.$$

Set the free variables $x_2 = p$, $x_4 = q$, $x_5 = r$. Then $x_3 = -2q - 3r$ and $x_1 = -2p - 2q - 5r$, so the full relation set is

$$x = (-2p - 2q - 5r, p, -2q - 3r, q, r)^\top, \quad p, q, r \in \mathbb{R}.$$

Thus redundant generators create nonunique coefficients, while a_1, a_3 give unique coefficients in their generated plane.

4. Vector spaces and subspaces

A **real vector space** is a set V with addition of its elements and multiplication by real scalars. Both operations take their values in V . They satisfy the following eight axioms, for all $u, v, w \in V$ and $a, b \in \mathbb{R}$:

- VS 1. $(u + v) + w = u + (v + w)$,
- VS 2. There is $0_V \in V$ with $0_V + u = u + 0_V = u$,
- VS 3. $u + (-1)u = 0_V$,
- VS 4. $u + v = v + u$,
- VS 5. $a(u + v) = au + av$,
- VS 6. $(a + b)u = au + bu$,
- VS 7. $(ab)u = a(bu)$,
- VS 8. $1u = u$.

Elements of V are called **vectors**, even when they are matrices or functions. Write $-v = (-1)v$ and $u - v = u + (-v)$. We usually write 0 for 0_V when the ambient space is specified. Associativity and commutativity allow regrouping finite sums. For example,

$$((u + v) + w) + z = (u + (v + w)) + z = ((v + w) + u) + z = (v + w) + (u + z).$$

Proposition 2.7 (Consequences of the axioms). The zero vector and each additive inverse are unique. Cancellation holds: $u + v = u + w$ implies $v = w$. Moreover, $0v = 0_V$, $a0_V = 0_V$, and $av = 0_V$ implies $a = 0$ or $v = 0_V$.

Proof. If $0'_V$ is another zero, then $0'_V = 0'_V + 0_V = 0_V$. For cancellation, add $-u$ to both sides and regroup:

$$(-u) + (u + v) = ((-u) + u) + v = v, \quad (-u) + (u + w) = w.$$

Hence $v = w$. If $v + w = 0_V = v + (-v)$, cancellation gives $w = -v$; this proves uniqueness of the additive inverse. The axiom for $(-1)v$ gives

$$0_V = v + (-1)v = 1v + (-1)v = (1 - 1)v = 0v.$$

Also

$$a0_V = a(0_V + 0_V) = a0_V + a0_V.$$

Subtracting $a0_V$ from both sides gives $0_V = a0_V$. Finally, if $a \neq 0$ and $av = 0_V$, then

$$v = 1v = (a^{-1}a)v = a^{-1}(av) = a^{-1}0_V = 0_V.$$

□

Example 2.6 (Matrices and functions). The set $\mathbb{R}^{m \times n}$ is a vector space. Matrix addition and scaling are defined entrywise, so the real-number laws verify the eight axioms. For example,

$$((A + B) + C)_{ij} = (a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij}) = (A + (B + C))_{ij},$$

$$(a(A + B))_{ij} = a(a_{ij} + b_{ij}) = aa_{ij} + ab_{ij} = (aA + aB)_{ij}.$$

These identities hold for every entry. Its zero vector is the zero matrix, and the negative of A has entries $-a_{ij}$.

If T is any set, the real functions $f : T \rightarrow \mathbb{R}$ form a vector space under pointwise operations

$$(f + g)(t) = f(t) + g(t), \quad (af)(t) = af(t).$$

For every $t \in T$,

$$\begin{aligned} ((f + g) + h)(t) &= (f(t) + g(t)) + h(t) \\ &= f(t) + (g(t) + h(t)) = (f + (g + h))(t), \\ (a(f + g))(t) &= a(f(t) + g(t)) = af(t) + ag(t) = (af + ag)(t). \end{aligned}$$

The other axioms follow pointwise from the corresponding real-number laws. The zero vector is the function identically zero. A function which vanishes at some points but not at all points is not this zero vector; for example, $f(t) = t$ on $\mathbb{R}$ has $f(0) = 0$ but $f(1) = 1$.

A subset W of V is a **subspace** if it is a vector space with the same operations.

Theorem 2.8 (Subspace test). A subset $W \subseteq V$ is a subspace if and only if

- (a) $0_V \in W$;
- (b) $u + v \in W$ whenever $u, v \in W$;
- (c) $cu \in W$ whenever $u \in W$ and $c \in \mathbb{R}$.

Equivalently, W is nonempty and $au + bv \in W$ for all $u, v \in W$ and $a, b \in \mathbb{R}$.

Proof. A subspace satisfies these conditions by its own vector-space axioms. Conversely, (b,c) ensure the operations remain in W , (a) supplies its zero, and (c) supplies $-u = (-1)u \in W$. All the identities VS 1–VS 8 then hold in W because they already hold in V .

The three conditions imply $au + bv \in W$ by scaling twice and adding. Conversely, suppose the combined condition holds and choose $w \in W$. Then $0w + 0w = 0_V \in W$. For $u, v \in W$, taking $(a, b) = (1, 1)$ gives $u + v \in W$, and taking $(a, b) = (c, 0)$ gives $cu \in W$. $\square$

Example 2.7 (A coordinate restriction and perpendicularity). The set $W = \{x \in \mathbb{R}^n : x_n = 0\}$ is a subspace: its zero vector belongs to W , and

$$(x + y)_n = x_n + y_n = 0, \quad (cx)_n = cx_n = 0$$

for $x, y \in W$. For $n \geq 2$, deleting the last, zero coordinate identifies its addition and scaling with those of $\mathbb{R}^{n-1}$.

For fixed $a \in \mathbb{R}^n$, let $U = \{x : x \cdot a = 0\}$. The three checks are

$$0 \cdot a = 0, \quad (x + y) \cdot a = x \cdot a + y \cdot a = 0, \quad (cx) \cdot a = c(x \cdot a) = 0.$$

Thus the vectors perpendicular to a form a subspace, including when $a = 0$ and the subspace is all of $\mathbb{R}^n$.

Example 2.8 (Tests that distinguish linear and nonlinear restrictions). The solution set of $Ax = 0$ is a subspace, since $A(au + bv) = aAu + bAv = 0$. For fixed $b \neq 0$, a nonempty solution set of $Ax = b$ is not a subspace, since it does not contain 0. Symmetric matrices form a subspace: zero is symmetric, and for symmetric S, T ,

$$(S + T)^\top = S^\top + T^\top = S + T, \quad (cS)^\top = cS^\top = cS.$$

Invertible matrices do not: the zero matrix is excluded. The set $\{(x, y) : xy = 0\}$ contains both coordinate axes but fails addition: $(1, 0) + (0, 1) = (1, 1)$ is outside it.

Polynomials of degree at most d , including the zero polynomial, form a subspace of real functions. Polynomials of degree exactly d do not. On a fixed open interval, continuous functions form a subspace: zero is continuous, and sums and scalar multiples of continuous functions are continuous. Differentiable functions likewise form a subspace, since

$$(f + g)' = f' + g', \quad (cf)' = cf', \quad 0' = 0.$$

Every differentiable function is continuous, so the differentiable functions also form a subspace of the continuous functions.

For subspaces $U, W \subseteq V$, their **intersection** consists of vectors belonging to both, and their **sum** consists of sums of one vector from each:

$$U \cap W = \{v : v \in U \text{ and } v \in W\}, \quad U + W = \{u + w : u \in U, w \in W\}.$$

The intersection is a subspace. It contains zero. If $x, y \in U \cap W$, then $x + y$ belongs to both U, W , as does cx for every scalar c . The sum is also a subspace: $0 = 0 + 0$ belongs to it, and

$$(u_1 + w_1) + (u_2 + w_2) = (u_1 + u_2) + (w_1 + w_2) \in U + W, \\ c(u_1 + w_1) = cu_1 + cw_1 \in U + W.$$

For example, the coordinate planes $z = 0$ and $y = 0$ in $\mathbb{R}^3$ intersect in $\{(t, 0, 0) : t \in \mathbb{R}\}$, the x -axis. Their sum is all of $\mathbb{R}^3$, because $(x, y, z) = (x, y, 0) + (0, 0, z)$.

The **union** $U \cup W$ consists of vectors lying in at least one of the two subspaces. It need not be a subspace. More precisely, $U \cup W$ is a subspace exactly when one of U, W contains the other. To prove necessity, if neither contains the other, choose $u \in U \setminus W$ and $w \in W \setminus U$. The vector $u + w$ belongs to neither set: membership in U would force $w = (u + w) - u \in U$, and membership in W would force $u \in W$. Sufficiency follows because the union is then the larger subspace.

5. Span and affine geometry

Vectors $v_1, \dots, v_k$ **generate** a vector space V if each $v \in V$ has an expression $v = x_1v_1 + \dots + x_kv_k$ with real coefficients. Their **span** is the set of all these combinations:

$$\text{span}(v_1, \dots, v_k) = \{x_1v_1 + \dots + x_kv_k : x_1, \dots, x_k \in \mathbb{R}\}.$$

Proposition 2.9 (The subspace generated by a list). The span of a finite list is a subspace. It is the smallest subspace containing every vector in that list.

Proof. Zero belongs to the span because $0 = 0v_1 + \dots + 0v_k$. Given two of its elements $u = \sum_i x_i v_i$ and $w = \sum_i y_i v_i$,

$$\begin{aligned} u + w &= (x_1v_1 + \dots + x_kv_k) + (y_1v_1 + \dots + y_kv_k) \\ &= (x_1 + y_1)v_1 + \dots + (x_k + y_k)v_k, \\ cu &= (cx_1)v_1 + \dots + (cx_k)v_k. \end{aligned}$$

Both expressions are combinations of the same list, proving closure. Every v_i belongs to the span by giving it coefficient one and all other vectors coefficient zero. If a subspace Z contains each v_i , then closure under scaling puts every $x_i v_i$ in Z , and repeated addition puts $\sum_i x_i v_i$ in Z . Thus the span is contained in any such Z . $\square$

For an arbitrary set S , define $\text{span}(S)$ using all *finite* combinations of its elements. The sum of two finite combinations is a finite combination formed by joining their lists; scaling multiplies their coefficients. The preceding proof therefore also gives the subspace and minimality claims for S . The empty combination is zero, so $\text{span}(\emptyset) = \{0\}$.

For rectangular matrices, let $M_{rs} = e_r e_s^\top$, where $e_r \in \mathbb{R}^m$ and $e_s \in \mathbb{R}^n$; thus M_{rs} has its single nonzero entry, equal to 1, in position (r, s) . The standard vectors generate $\mathbb{R}^n$, these M_{rs} generate $\mathbb{R}^{m \times n}$, and $1, t, \dots, t^d$ generate the polynomials of degree at most d . These statements follow from

$$x = \sum_i x_i e_i, \quad A = \sum_{r,s} a_{rs} M_{rs}, \quad p(t) = \sum_{j=0}^d c_j t^j.$$

Two different lists can have the same span, as in Example 2.5. Equality of spans is proved by expressing each list in terms of the other, not by requiring the lists themselves to be equal.

Example 2.9 (Worked reading: a matrix-defined subspace). Fix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. The matrices commuting with A form a subspace: if $AB = BA$ and $AC = CA$, then $A(sB + tC) = (sB + tC)A$. To find all of them, write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The two products are

$$AB = \begin{pmatrix} a + 2c & b + 2d \\ 3a + 4c & 3b + 4d \end{pmatrix}, \quad BA = \begin{pmatrix} a + 3b & 2a + 4b \\ c + 3d & 2c + 4d \end{pmatrix}.$$

Equating their $(1, 1), (1, 2), (2, 1)$ entries gives

$$2c = 3b, \quad 2d = 2a + 3b, \quad d = a + c.$$

The $(2, 2)$ entries repeat $3b = 2c$. The third equation follows from the first two. Thus $c = 3b/2$, $d = a + 3b/2$, and the full subspace is

$$B = aI + \frac{b}{2} \begin{pmatrix} 0 & 2 \\ 3 & 3 \end{pmatrix}.$$

Conversely, both displayed generating matrices commute with A , so every matrix in this span satisfies the original equation.

For a nonzero vector $u \in V$, the line through zero in direction u is the subspace $\{tu : t \in \mathbb{R}\}$. If neither of u, v is a scalar multiple of the other, their span

$$\{su + tv : s, t \in \mathbb{R}\}$$

is the plane through zero generated by u, v . For a subset $S \subseteq V$ and a vector p , its **translate** is $p + S = \{p + s : s \in S\}$. In particular,

$$p + tu \quad (t \in \mathbb{R}), \quad p + su + tv \quad (s, t \in \mathbb{R})$$

describe the parallel line and plane passing through p .

For a subspace W , its translate $p + W$ is a subspace exactly when $p \in W$. If $p \in W$, then $p + w \in W$ for every $w \in W$; conversely, every $w \in W$ equals $p + (w - p) \in p + W$. Thus $p + W = W$. If $p + W$ is a subspace, it contains zero, so $p + w = 0$ for some $w \in W$. Then $p = -w \in W$, proving the converse. If $Ap = b$, then

$$Ax = b \iff A(x - p) = 0 \iff x = p + z \text{ with } Az = 0.$$

Thus the whole solution set is a translate of a subspace, with the homogeneous equation determining its directions.

The **segment** from v to $v + u$ is $\{v + tu : 0 \leq t \leq 1\}$. The endpoints occur at $t = 0, 1$; $t = 1/2$ gives the midpoint and $t = 1/3$ gives the point one third of the way from v to $v + u$. For endpoints v, w , substitute $u = w - v$:

$$v + t(w - v) = v + tw - tv = (1 - t)v + tw.$$

Consequently the segment has the two descriptions

$$[v, w] = \{(1 - t)v + tw : 0 \leq t \leq 1\} = \{sv + tw : s, t \geq 0, s + t = 1\}.$$

For the forward inclusion, put $s = 1 - t$; then $s \geq 0$ and $s + t = 1$. For the reverse inclusion, $s, t \geq 0$ and $s + t = 1$ imply $0 \leq t \leq 1$ and $s = 1 - t$, giving the first form. These descriptions include $v = w$, when the segment is one point.

More generally, coefficients whose sum is one give an **affine combination**; if they are also nonnegative, they give a **convex combination**. For $w_i \in W$ and $\sum_i c_i = 1$,

$$\sum_i c_i(p + w_i) = \left(\sum_i c_i\right)p + \sum_i c_i w_i = p + \sum_i c_i w_i \in p + W.$$

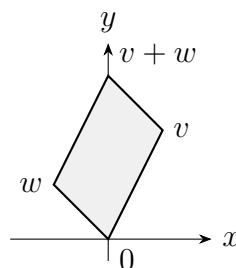
Thus affine combinations of points in a translate $p + W$ remain in that translate.

Example 2.10 (A parallelogram through coefficient inequalities). With $v = (1, 2)^\top$ and $w = (-1, 1)^\top$, the parallelogram they span is

$$\{sv + tw : 0 \leq s, t \leq 1\}.$$

Its vertices are $0, v, v + w = (0, 3)^\top, w$. If $(x, y)^\top = sv + tw$, then $x = s - t, y = 2s + t$, so $s = (x + y)/3$ and $t = (y - 2x)/3$. An equivalent description is therefore

$$0 \leq x + y \leq 3, \quad 0 \leq y - 2x \leq 3.$$



The segment between v and w has the extra equation $s + t = 1$. The parallelogram allows the two coefficients to vary independently. Its translate by p consists of $p + sv + tw$ with the same bounds.

Three vectors similarly generate a box $\{t_1v_1 + t_2v_2 + t_3v_3 : 0 \leq t_i \leq 1\}$; adding p translates the box. Such a set may collapse into a plane, a segment, or a point when the generating vectors satisfy relations. The coefficient description remains valid in these degenerate cases.

Exercises and solutions

Suggested practice: 2.2, 2.6, 2.10, 2.15, 2.17, 2.25, 2.28, 2.30, 2.35, 2.36.

Elementary operations and inverses

Exercise 2.1 (Elementary row operations). For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$, find the elementary matrix and the resulting product for each operation, applied separately: swap rows 1, 3; multiply row 2 by 7; add 5 times row 2 to row 3.

Solution. The three matrices and products are

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 5 & 6 \\ 3 & 4 \\ 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 \\ 21 & 28 \\ 5 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 20 & 26 \end{pmatrix}.$$

For example, the third product has last row $(5, 6) + 5(3, 4) = (20, 26)$. These are separate operations on the original A .

Exercise 2.2 (* Outer-product formulas). Derive formulas in terms of coordinate vectors for a row swap, a nonzero row scaling, and addition of a multiple of one row to another.

Solution. Since $e_s^\top x = x_s$, the vector $e_r e_s^\top x = x_s e_r$ places coordinate s of x in position r and zero elsewhere. For a swap, $r \neq s$, the required result is

$$x + (x_s - x_r)e_r + (x_r - x_s)e_s = x - (e_r - e_s)(e_r - e_s)^\top x.$$

Thus $S_{rs} = I - (e_r - e_s)(e_r - e_s)^\top$. Scaling coordinate r by $c \neq 0$ gives

$$x + (c - 1)x_r e_r = [I + (c - 1)e_r e_r^\top]x,$$

so $D_r(c) = I + (c - 1)e_r e_r^\top$. Adding c times coordinate s to coordinate r gives

$$x + cx_s e_r = [I + ce_r e_s^\top]x,$$

so $T_{rs}(c) = I + ce_r e_s^\top$, $r \neq s$. Since the formulas hold for every column, left multiplication applies the respective operation to every column of a matrix, hence to its rows.

Exercise 2.3 (Transposes and inverses). For the three elementary matrices S_{rs} , $D_r(c)$, $T_{rs}(c)$, compute their transposes and inverses. Are both again elementary?

Solution. Since $(e_r e_s^\top)^\top = e_s e_r^\top$,

$$S_{rs}^\top = S_{rs}, \quad D_r(c)^\top = D_r(c), \quad T_{rs}(c)^\top = T_{sr}(c).$$

The inverses are S_{rs} , $D_r(c^{-1})$, and $T_{rs}(-c)$. For the swap, applying it twice fixes every coordinate. Scaling by c then by c^{-1} restores that row. For row addition, $(I + ce_r e_s^\top)(I - ce_r e_s^\top) = I$, since $e_s^\top e_r = 0$. Reversing the product gives the same identity. All are elementary; a zero row-addition coefficient gives the identity operation.

Exercise 2.4 (Products and operation order). Is a product of two elementary matrices always elementary? For $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ compare, in both orders, the swap $R_1 \leftrightarrow R_2$ with (i) $R_2 \leftarrow 5R_2$ and (ii) $R_2 \leftarrow R_2 + 3R_1$. Do elementary matrices commute?

Solution. The product $\begin{pmatrix} 0 & 1 \\ 5 & 0 \end{pmatrix} = D_2(5)S_{12}$ is not elementary: it is neither a pure swap, a diagonal scaling, nor an identity with one off-diagonal change. For (i), the two products are

$$D_2(5)S_{12} = \begin{pmatrix} 0 & 1 \\ 5 & 0 \end{pmatrix}, \quad S_{12}D_2(5) = \begin{pmatrix} 0 & 5 \\ 1 & 0 \end{pmatrix}.$$

They give respectively $\begin{pmatrix} 4 & 5 & 6 \\ 5 & 10 & 15 \end{pmatrix}$ and $\begin{pmatrix} 20 & 25 & 30 \\ 1 & 2 & 3 \end{pmatrix}$. For (ii), the two operation matrices are

$$T_{21}(3)S_{12} = \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}, \quad S_{12}T_{21}(3) = \begin{pmatrix} 3 & 1 \\ 1 & 0 \end{pmatrix}.$$

They give respectively $\begin{pmatrix} 4 & 5 & 6 \\ 13 & 17 & 21 \end{pmatrix}$ and $\begin{pmatrix} 7 & 11 & 15 \\ 1 & 2 & 3 \end{pmatrix}$. Thus elementary matrices generally do not commute.

Exercise 2.5 (A table of elementary checks). Compute the transpose and inverse of the following 3×3 matrices: I , S_{23} , $D_1(5)$, $T_{21}(4)$, $T_{12}(4)$, $T_{21}(0)$.

Solution. Their entries, transposes and inverses are completely specified by

E	$E^\top$	E^{-1}
I	I	I
S_{23}	S_{23}	S_{23}
$\text{diag}(5, 1, 1)$	$\text{diag}(5, 1, 1)$	$\text{diag}(1/5, 1, 1)$
$I + 4e_2e_1^\top$	$I + 4e_1e_2^\top$	$I - 4e_2e_1^\top$
$I + 4e_1e_2^\top$	$I + 4e_2e_1^\top$	$I - 4e_1e_2^\top$
I	I	I

Here $S_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. In the two row-addition cases the off-diagonal matrix squares to zero, so multiplying the displayed E and E^{-1} cancels the linear terms and leaves I . The other products follow directly from the diagonal or swap entries.

Exercise 2.6 (Column operations). Prove that elementary column operations can be represented by right multiplication. For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$, give the right multiplier and result when, separately, the columns are swapped, column 1 is multiplied by 3, and 5 times column 1 is added to column 2.

Solution. Transpose A . A column operation on A is a row operation $EA^\top$ on $A^\top$. Transposition back gives $AE^\top$, proving the assertion. The multipliers are

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix},$$

and the products are

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{pmatrix}, \quad \begin{pmatrix} 3 & 2 \\ 9 & 4 \\ 15 & 6 \end{pmatrix}, \quad \begin{pmatrix} 1 & 7 \\ 3 & 19 \\ 5 & 31 \end{pmatrix}.$$

For the last operation the multiplier has its 5 in position (1, 2).

Exercise 2.7 (* Recording a complete elimination). Show that every $m \times n$ matrix A admits an equation $E_k \cdots E_1 A = H$, where each E_i is elementary and H is in echelon form. The nonzero pivots may all be normalized to 1.

Solution. Start with the entire matrix as the active rectangle. If it is zero, stop. Otherwise find its first nonzero column, swap a row with a nonzero entry in that column to the top of the active rectangle, and multiply this row by the reciprocal of that entry. Subtract multiples of this pivot row from the rows below it. Continue with the rows below the pivot and the columns to its right. The active row count decreases at each stage, so the process stops after at most $\min(m, n)$ pivot stages. Earlier pivot columns remain zero below their pivots because every later active row has zeros there. Thus the final pivot columns strictly increase and all zero rows are at the bottom: the final matrix is echelon. Represent each recorded operation by its elementary matrix. If the chronological list is $E_1, \dots, E_k$, the final matrix is $E_k \cdots E_1 A$. Every factor is invertible, including the normalizations since their pivots are nonzero.

Exercise 2.8 (Row reduction followed by column reduction). Reduce $A = \begin{pmatrix} 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -2 & 0 & 4 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix}$ to echelon form by elementary row operations. Then use column operations to reduce that echelon matrix to $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

Solution. First swap rows 1, 2, then subtract the new row 1 from row 3:

$$A \xrightarrow{S_{12}} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix} \xrightarrow{T_{31}(-1)} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix}.$$

Rows 2, 3 now coincide. Subtract row 2 from row 3, then halve row 1. These operations, $T_{32}(-1)$ and $D_1(1/2)$, give

$$H = \begin{pmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The row multiplier is $D_1(1/2)T_{32}(-1)T_{31}(-1)S_{12}$. Swap columns 2, 3 of H to get

$$\begin{pmatrix} 1 & -1 & 2 & 0 & 2 \\ 0 & 1 & 0 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Now perform $C_2 \leftarrow C_2 + C_1$, $C_3 \leftarrow C_3 - 2C_1$, $C_5 \leftarrow C_5 - 2C_1$; this gives $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. Finally $C_4 \leftarrow C_4 - 2C_2$ and $C_5 \leftarrow C_5 - 3C_2$ give the requested matrix. All operations are reversible.

Exercise 2.9 (Recovering the accumulated multiplier). How can one find the matrix E in $EA = H$ while reducing A to echelon form?

Solution. Append I_m to A and perform each operation on both blocks. If the accumulated multiplier is E , the blocks are $[EA \mid EI_m] = [H \mid E]$. Stop when the left block has the desired form. For the preceding exercise, the same row operations change the appended identity as follows:

$$I_3 \xrightarrow{S_{12}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{T_{31}(-1)} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix},$$

$$\xrightarrow{T_{32}(-1)} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix} \xrightarrow{D_1(1/2)} E = \begin{pmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}.$$

Thus the final augmented matrix is $[H \mid E]$. To verify $EA = H$, write the original rows of A as a_1, a_2, a_3 . Its product with E has rows

$$\frac{1}{2}a_2 = (1, 2, -1, 0, 2), \quad a_1 = (0, 0, 1, 2, 3), \quad a_3 - a_1 - a_2 = (0, 0, 0, 0, 0),$$

which are exactly the rows of H .

Exercise 2.10 (An inverse by elimination). Find the inverse of $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & 4 & 3 \end{pmatrix}$ by augmented row reduction.

Solution. Swapping rows 1, 2 of $[A \mid I]$ and subtracting $3R_1$ from R_3 gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \end{array} \right].$$

Now subtract R_2 from R_3 , obtaining $(0, 0, -1 \mid -1, -3, 1)$, and negate this row:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right].$$

Subtract R_3 from R_1 and R_2 , then subtract R_2 from R_1 :

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right].$$

Hence $A^{-1} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -3 & 1 \\ 1 & 3 & -1 \end{pmatrix}$. For a direct check, original row 1 uses inverse rows $2 + 3 = (1, 0, 0)$; row 2 uses rows $1 + 2 + 3 = (0, 1, 0)$; row 3 uses $3R_1 + 4R_2 + 3R_3 = (0, 0, 1)$. Thus $AA^{-1} = I$.

Exercise 2.11 (Two more inverse calculations). Find the inverses of $A = \begin{pmatrix} 2 & 1 & 2 \\ 0 & 3 & -1 \\ 4 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & 7 \end{pmatrix}$.

Solution. For $Ax = b$, the second equation gives $x_3 = 3x_2 - b_2$. The first then gives $2x_1 + 7x_2 = b_1 + 2b_2$, while the third gives $4x_1 + 4x_2 = b_3 + b_2$. Subtracting twice the first of these equations from the second gives $-10x_2 = b_3 - 2b_1 - 3b_2$. Consequently

$$\begin{aligned} x_2 &= \frac{2b_1 + 3b_2 - b_3}{10}, \\ x_3 &= 3x_2 - b_2 = \frac{6b_1 - b_2 - 3b_3}{10}, \\ x_1 &= \frac{b_1 + 2b_2 - 7x_2}{2} = \frac{-4b_1 - b_2 + 7b_3}{20}. \end{aligned}$$

Reading off the coefficients of b_1, b_2, b_3 in each coordinate gives

$$A^{-1} = \frac{1}{20} \begin{pmatrix} -4 & -1 & 7 \\ 4 & 6 & -2 \\ 12 & -2 & -6 \end{pmatrix}.$$

For $Bx = b$, subtract twice equation 2 from equation 3 to obtain $5x_3 = b_3 - 2b_2$. Then $x_2 = (7b_2 - b_3)/5$ and $x_1 = b_1 - 2x_2 + x_3 = b_1 - 16b_2/5 + 3b_3/5$. Hence

$$B^{-1} = \frac{1}{5} \begin{pmatrix} 5 & -16 & 3 \\ 0 & 7 & -1 \\ 0 & -2 & 1 \end{pmatrix}.$$

Both calculations give a unique solution for every b , proving that the displayed matrices are the inverses.

Exercise 2.12 (Existence in order two). Decide whether $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ have inverses; find any that exist.

Solution. $A(-2, 1)^\top = 0$ with $(-2, 1)^\top \neq 0$. If A^{-1} existed, multiplication by it would give $(-2, 1)^\top = A^{-1}0 = 0$, a contradiction. For B , the two-by-two inverse formula gives the candidate $C = \begin{pmatrix} -2 & 3/2 \\ 1 & -1/2 \end{pmatrix}$, since $1 \cdot 4 - 3 \cdot 2 = -2 \neq 0$. Directly,

$$BC = \begin{pmatrix} -2+3 & 3/2-3/2 \\ -4+4 & 3-2 \end{pmatrix} = I, \quad CB = \begin{pmatrix} -2+3 & -6+6 \\ 1-1 & 3-2 \end{pmatrix} = I.$$

Thus $C = B^{-1}$.

Exercise 2.13 (Left and right inverses). Suppose $BA = I$ and $AC = I$, with compatible sizes. Show that $B = C$. Deduce uniqueness of a two-sided inverse.

Solution. Associativity gives $B = B(AC) = (BA)C = C$. If B and C are both two-sided inverses, these two identities are among their defining identities, so the same proof applies. Even with rectangular A , the hypotheses force the two candidate inverses to coincide; no cancellation of a noninvertible matrix was used.

Exercise 2.14 (Scalar multiples and commuting inverses). For invertible A , prove $(cA)^{-1} = c^{-1}A^{-1}$ for $c \neq 0$ and $(A^{-1})^{-1} = A$. If $AB = BA$ and B is invertible, prove $AB^{-1} = B^{-1}A$.

Solution. The two products $(cA)(c^{-1}A^{-1})$ and $(c^{-1}A^{-1})(cA)$ equal I . The two identities $AA^{-1} = A^{-1}A = I$ also say precisely that A is an inverse of A^{-1} . Finally, multiply $AB = BA$ on the left and right by B^{-1} to obtain $B^{-1}A = AB^{-1}$. The last assertion does not require A invertible.

Exercise 2.15 (One-sided and rectangular inverses). (a) For square A, B of the same size, prove $AB = I$ if and only if $BA = I$.

(b) If A is $m \times n$ with $m < n$, prove that no $m \times n$ matrix B satisfies $B^T A = I_n$.

(c) Give a rectangular A with $AA^T = I$ but $A^T A \neq I$, and one with the two roles reversed.

Solution. (a) If $AB = I$, the equation $Bx = 0$ gives $x = ABx = 0$, so B is invertible by the square criterion. Thus $A = B^{-1}$ and $BA = I$. Interchanging A, B proves the converse. (b) A homogeneous system of $m < n$ equations has a free variable after elimination and hence has a nonzero solution $Ax = 0$. But $B^T A = I_n$ would give $x = B^T Ax = 0$, a contradiction. (c) Take $A = (1 \ 0)$: $AA^T = 1$, whereas $A^T A = \text{diag}(1, 0)$. Its transpose supplies the second example. Thus equal square sizes are essential in (a).

Exercise 2.16 (Orthogonal matrices). A real square matrix Q is called orthogonal when $Q^T Q = QQ^T = I$. Prove that Q is invertible, $Q^{-1} = Q^T$, Q^T is orthogonal, and the product of two orthogonal matrices is orthogonal.

Solution. The defining two equations make Q^T the inverse of Q . For Q^T , the two defining products are QQ^T and $Q^T Q$, both I . If P is also orthogonal, then $(QP)^T QP = P^T Q^T QP = P^T P = I$, and $QP(QP)^T = QPP^T Q^T = QQ^T = I$.

Exercise 2.17 (An outer-product update of an inverse). Let A be invertible and $a, b \in \mathbb{R}^n$. Prove

$$(A + ab^T)^{-1} = A^{-1} - \frac{A^{-1}ab^T A^{-1}}{1 + b^T A^{-1}a} \quad \text{if } 1 + b^T A^{-1}a \neq 0.$$

Deduce the formulas for $(A + aa^T)^{-1}$ and $(A - aa^T)^{-1}$, stating their respective nonzero denominators.

Solution. Put $s = b^T A^{-1}a$ and $C = A^{-1} - (1 + s)^{-1}A^{-1}ab^T A^{-1}$. Expansion gives

$$(A + ab^T)C = I + \left(1 - \frac{1}{1 + s} - \frac{s}{1 + s}\right)ab^T A^{-1} = I.$$

In this multiplication the last term reduces because $ab^T A^{-1}ab^T A^{-1} = s ab^T A^{-1}$. In the reverse order, the corresponding reduction is $A^{-1}ab^T A^{-1}ab^T = s A^{-1}ab^T$, giving

$$C(A + ab^T) = I + \left(1 - \frac{1}{1 + s} - \frac{s}{1 + s}\right)A^{-1}ab^T = I.$$

For $b = a$ this gives $A^{-1} - A^{-1}aa^T A^{-1}/(1 + a^T A^{-1}a)$, requiring $1 + a^T A^{-1}a \neq 0$. For $b = -a$ it gives $A^{-1} + A^{-1}aa^T A^{-1}/(1 - a^T A^{-1}a)$, requiring $1 - a^T A^{-1}a \neq 0$. Symmetry of A was not assumed.

Exercise 2.18 (Integer restrictions on a solution line). A farmer buys 100 animals for 100 crowns. Sheep cost $1/2$ crown, goats $4/3$ crowns, and hogs $7/2$ crowns. Find all possible nonnegative integer counts. Which solution has eleven more sheep than goats?

Solution. Writing the counts as (x, y, z) gives $x + y + z = 100$ and $x/2 + 4y/3 + 7z/2 = 100$. Subtract half the first equation from the second: $5y + 18z = 300$. Thus

$$(x, y, z) = (40, 60, 0) + z(13/5, -18/5, 1).$$

Nonnegativity requires $0 \leq z \leq 50/3$. Since $18z$ must be divisible by 5, z is a multiple of 5. The four possibilities are $(40, 60, 0)$, $(53, 42, 5)$, $(66, 24, 10)$ and $(79, 6, 15)$. The extra equation $x - y = 11$ becomes $-20 + 31z/5 = 11$, so $z = 5$ and the counts are $(53, 42, 5)$.

Exercise 2.19 (Two Gaussian eliminations). Solve the systems with augmented matrices

$$\left[\begin{array}{ccc|c} 2 & 1 & -3 & 1 \\ 5 & 2 & -6 & 5 \\ 3 & -1 & -4 & 7 \end{array} \right], \quad \left[\begin{array}{ccc|c} 2 & 1 & -2 & 10 \\ 3 & 2 & 2 & 1 \\ 5 & 4 & 3 & 4 \end{array} \right].$$

Solution. For the first, use $R_2 \leftarrow 2R_2 - 5R_1$ and $R_3 \leftarrow 2R_3 - 3R_1$ to obtain rows $(0, -1, 3 \mid 5)$ and $(0, -5, 1 \mid 11)$. Then $R_3 \leftarrow R_3 - 5R_2$ gives $(0, 0, -14 \mid -14)$. Thus $x_3 = 1$, $x_2 = -2$, $x_1 = 3$. For the second, use $R_2 \leftarrow 2R_2 - 3R_1$ and $R_3 \leftarrow 2R_3 - 5R_1$ to get $(0, 1, 10 \mid -28)$ and $(0, 3, 16 \mid -42)$. Then $R_3 \leftarrow R_3 - 3R_2$ gives $(0, 0, -14 \mid 42)$. Thus $x_3 = -3$, $x_2 = 2$, $x_1 = 1$. Each combined row replacement used above is a nonzero scaling followed by addition, hence reversible. Both systems have three pivots.

Exercise 2.20 (Columns of a product). If A is $m \times n$ and B is $n \times p$, show that column k of AB is $\sum_{j=1}^n b_{jk}a_j$, where a_j is column j of A . If $AB = C$, which vector solves $Ax = c_k$?

Solution. The i th entry of column k is $(AB)_{ik} = \sum_j a_{ij}b_{jk}$, which is the i th entry of the claimed combination. If b_k is column k of B , then $Ab_k = c_k$; thus $x = b_k$ is a solution. All columns of $AB = C$ are consequently separate systems sharing the coefficient matrix A . Uniqueness requires additional hypotheses, such as square invertibility of A .

Vector spaces, subspaces, and span

Exercise 2.21 (Zero and familiar spaces). Explain why every vector space is nonempty and why $\{0\}$ is itself a vector space. Verify that $\mathbb{R}$, $\mathbb{R}^m$, and $\mathbb{R}^{m \times n}$ are real vector spaces, identifying the vectors and zero in each case.

Solution. The zero axiom requires an element 0, so a vector space is nonempty. On $\{0\}$, both operations always give 0: every axiom has both sides 0, the additive inverse is 0, and $1 \cdot 0 = 0$. In $\mathbb{R}$, vectors are numbers and zero is the number 0; all axioms are the real-number laws. In $\mathbb{R}^m$ vectors are m -tuples, with zero the tuple of zeros. In $\mathbb{R}^{m \times n}$ they are matrices, with zero the zero matrix. Addition and scaling in the last two cases are coordinatewise. Each coordinate of the commutative, associative and distributive identities is the corresponding real-number identity; additive inverses negate each coordinate, scalar associativity is $(ab)x_i = a(bx_i)$, and $1x_i = x_i$. Closure follows because every resulting coordinate is real and the sizes stay fixed.

Exercise 2.22 (* Addition from the axioms). Prove that a finite sum in a vector space is independent of parenthesization, that the zero vector and each additive inverse are unique, and that $x + y = x + z$ implies $y = z$.

Solution. Define $s_1 = x_1$ and $s_n = s_{n-1} + x_n$. By induction and associativity, $s_n = s_j + (x_{j+1} + \cdots + x_n)$ for every $1 \leq j < n$, where the second sum is left-associated. Any parenthesized expression has an outer split after some j . Induction on its two shorter expressions makes them the corresponding left-associated sums; the split identity then makes their sum s_n . This proves independence of parenthesization. If o, o' both act as zero, $o = o + o' = o'$. If $x + u = x + v = 0$, then $u = u + (x + v) = (u + x) + v = v$. Finally, adding $-x$ to $x + y = x + z$ gives $((-x) + x) + y = ((-x) + x) + z$, hence $y = z$.

Exercise 2.23 (Scalar multiplication from the axioms). Prove: $a0 = 0$; $0x = 0$; $ax = 0 \Rightarrow a = 0$ or $x = 0$; $(-a)x = a(-x) = -(ax)$; cancellation of a nonzero scalar; cancellation of a nonzero vector in $ax = bx$; and the two distributive subtraction identities.

Solution. From $a0 = a(0 + 0) = a0 + a0$, cancellation gives $a0 = 0$. Likewise $0x = (0 + 0)x = 0x + 0x$ gives $0x = 0$. If $a \neq 0$ and $ax = 0$, multiply by a^{-1} to get $x = 0$. Both $(-a)x$ and $a(-x)$ add to ax to give zero: $(a - a)x = 0$ and $a(x - x) = 0$. Uniqueness of additive inverses proves the claimed negative identities. If $ax = ay$ with $a \neq 0$, multiply by a^{-1} to get $x = y$. If $ax = bx$ with $x \neq 0$, then $(a - b)x = 0$, so $a = b$. Finally, $(a - b)x = ax + (-b)x = ax - bx$ and $a(x - y) = ax + a(-y) = ax - ay$.

Exercise 2.24 (The subspace test). Let W be a nonempty subset of a vector space, closed under addition and scalar multiplication. Prove that it is a vector space, that it contains zero, and identify the smallest and largest subspaces of V .

Solution. Choose $w \in W$. Closure under scaling gives $0 = 0w \in W$ and $-w = (-1)w \in W$. The two operations stay in W ; the associativity, commutativity, distributivity, scalar associativity and identity axioms are restrictions of those in V . Thus all vector-space axioms hold in W . Since every subspace contains 0, the smallest is $\{0\}$. Since every subspace is a subset of V and V satisfies the axioms, V is the largest.

Exercise 2.25 (All subspaces of the plane). Decide whether each subset of $\mathbb{R}^2$ is a subspace: $x = 0$; $x = 1$; $x \geq 0, y \geq 0$. Prove that every subspace of $\mathbb{R}^2$ is $\{0\}$, a line through zero, or $\mathbb{R}^2$.

Solution. The first is $\{(0, t) : t \in \mathbb{R}\}$, closed under linear combinations. The second excludes 0. The third contains $(1, 0)$ but not its negative. For the classification, let $W \neq \{0\}$ and choose a nonzero $a \in W$. It contains the line $\mathbb{R}a$. If it contains nothing else, it is that line. Otherwise choose $b \in W \setminus \mathbb{R}a$. Writing $a = (a_1, a_2)^\top$, $b = (b_1, b_2)^\top$, we have $a_1b_2 - a_2b_1 \neq 0$: if $a_1 \neq 0$, equality would give $b = (b_1/a_1)a$; if $a_1 = 0$, then $a_2 \neq 0$, equality forces $b_1 = 0$ and again $b \in \mathbb{R}a$. Therefore $(a \ b)$ is invertible by the two-by-two formula. Every $x \in \mathbb{R}^2$ equals $sa + tb$, and closure gives $x \in W$. Hence $W = \mathbb{R}^2$. Each of the three listed types does satisfy the subspace test.

Exercise 2.26 (Subsets of three-dimensional space). Which sets in $\mathbb{R}^3$ are subspaces: $x = 0$; $xy = 0$; $\{0\}$; the set of all combinations of two fixed vectors; and $ax + by + cz = 0$ for fixed real a, b, c ?

Solution. All except $xy = 0$ are subspaces. For $x = 0$, the first coordinate of any linear combination remains zero. For $\{0\}$ the test is immediate. For combinations $su + tv$, a combination of two such vectors has the same form after collecting coefficients. For $ax + by + cz = 0$, evaluating the left side on a linear combination gives the same combination of two zeros. This remains true if $a = b = c = 0$, when the set is all of $\mathbb{R}^3$. For $xy = 0$, the vectors $(1, 0, 0)$ and $(0, 1, 0)$ belong, but their sum $(1, 1, 0)$ does not. The set is a union of two planes, not one plane.

Exercise 2.27 (Matrix subspaces). Show that the lower triangular 3×3 matrices and the symmetric 3×3 matrices each form a subspace of $\mathbb{R}^{3 \times 3}$.

Solution. Both contain the zero matrix. In the triangular case, if $i < j$ then $(aA + bB)_{ij} = a \cdot 0 + b \cdot 0 = 0$, so the combination is lower triangular. In the symmetric case, $(aA + bB)^\top = aA^\top + bB^\top = aA + bB$. The subspace test applies in each case, with no condition on the other entries.

Exercise 2.28 (Intersection, union, and uniqueness of a sum). For subspaces $U, W \subseteq V$, prove that $U \cap W$ and $U + W$ are subspaces, whereas $U \cup W$ need not be. Prove that all decompositions $x = u + w$ in $U + W$ are unique when $U \cap W = \{0\}$; if the intersection contains a nonzero vector, show that some decompositions are not unique.

Solution. Zero belongs to the intersection and sum. If $x, y \in U \cap W$, then $ax + by$ belongs to each subspace. If $x = u_1 + w_1$ and $y = u_2 + w_2$, then $ax + by = (au_1 + bu_2) + (aw_1 + bw_2) \in U + W$. These prove the two subspace claims. The union of the two coordinate axes in $\mathbb{R}^2$ is not closed under addition because $e_1 + e_2$ belongs to neither axis. If $u_1 + w_1 = u_2 + w_2$, then $u_1 - u_2 = w_2 - w_1 \in U \cap W$. A zero intersection therefore gives $u_1 = u_2$ and $w_1 = w_2$. If $0 \neq z \in U \cap W$, the same vector z has the two decompositions $z + 0$ and $0 + z$. In fact every $u + w$ also equals $(u + z) + (w - z)$, giving distinct decompositions.

Exercise 2.29 (Finite combinations and the smallest subspace). Prove that a finite linear combination of elements of a subspace remains in it. For any subset $S \subseteq V$, prove that its span is a subspace and is the smallest subspace containing S .

Solution. Induction on the number of summands proves the first assertion: each scalar multiple lies in the subspace, and adding the next one preserves membership. The empty sum is 0. In the span, if $u = \sum_i a_i s_i$ and $v = \sum_j b_j t_j$, then $cu + dv = \sum_i ca_i s_i + \sum_j db_j t_j$ is again a finite combination of elements of S . Thus the span passes the subspace test, even when S is empty. It contains S using coefficient 1. Every subspace containing S contains each of its finite combinations by the first assertion, hence contains the span.

Exercise 2.30 (Geometrically different generating sets). Find the span of (a) a circle of positive radius in $\mathbb{R}^3$ centered at zero; (b) a plane in $\mathbb{R}^3$ not containing zero. Also determine the spans of $\{e_1, e_2, (0, -2, 0)^\top\}$, $\{e_1, e_2\}$, and $\{e_2, (0, -2, 0)^\top\}$.

Solution. (a) All circle points lie in its plane, so the span is contained there. Choose two perpendicular radius vectors u, v on the circle, of length $\rho > 0$, and use their directions as coordinate axes within the plane. A vector with planar coordinates (s, t) is $(s/\rho)u + (t/\rho)v$. Thus every vector of the plane is in the span of two circle points, proving the reverse inclusion. (b) Write the plane as $P = \{x : a^\top x = c\}$ with $a \neq 0$ and $c \neq 0$, and choose $p \in P$. Its direction plane is $W = \{w : a^\top w = 0\}$. For $w \in W$, both $p + w$ and p lie in P , so their difference w lies in its span. Any $x \in \mathbb{R}^3$ can be written $x = (a^\top x/c)p + [x - (a^\top x/c)p]$, with the bracket in W . Thus $\text{span}(P) = \mathbb{R}^3$. Finally, $(0, -2, 0)^\top = -2e_2$. The first two lists therefore generate exactly $\{(s, t, 0)^\top : s, t \in \mathbb{R}\}$, the xy -plane. The last list generates exactly $\{(0, t, 0)^\top : t \in \mathbb{R}\}$, the y -axis.

Exercise 2.31 (Orthogonality constraints). For fixed $a_1, \dots, a_r \in \mathbb{R}^n$, show that $W = \{x : x^\top a_i = 0 \text{ for every } i\}$ is a subspace. For a subspace $V \subseteq \mathbb{R}^n$, prove the same for $V^\perp = \{x : x^\top v = 0 \text{ for every } v \in V\}$.

Solution. Zero satisfies every constraint. If $x, y \in W$, then for every i , $(sx + ty)^\top a_i = sx^\top a_i + ty^\top a_i = 0$. Thus W is a subspace. The identical argument with an arbitrary $v \in V$ gives $(sx + ty)^\top v = 0$ whenever $x, y \in V^\perp$. The quantifier “every” is preserved because the calculation holds for each chosen v , so the subspace test also proves the second assertion.

Exercise 2.32 (Equations defining subspaces). Prove that the following sets are subspaces: in $\mathbb{R}^2$, $x = y$, $x - y = 0$, and $x + 4y = 0$; in $\mathbb{R}^3$, $x + y + z = 0$, the simultaneous conditions $x = y$, $2y = z$, and $x + y = 3z$.

Solution. The first two sets coincide and equal $\text{span}((1, 1)^\top)$. The third equals $\text{span}((-4, 1)^\top)$. For the three sets in $\mathbb{R}^3$, solving their equations gives respectively

$$(x, y, z) = y(-1, 1, 0) + z(-1, 0, 1),$$

$$(x, y, z) = y(1, 1, 2),$$

$$(x, y, z) = y(-1, 1, 0) + z(3, 0, 1).$$

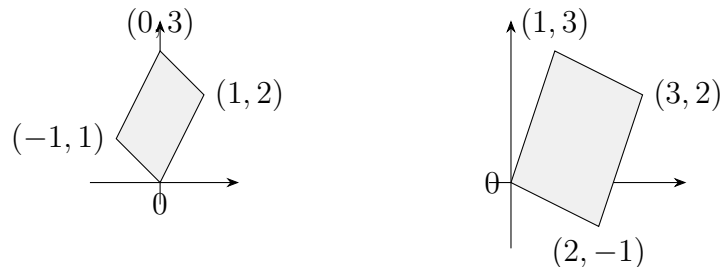
In each display the parameters are unrestricted real numbers and every listed combination satisfies the original equations. Each set is therefore a span and hence a subspace.

Exercise 2.33 (Testing orthogonality on generators). Suppose $a_1, \dots, a_r$ generate a subspace $V \subseteq \mathbb{R}^n$. Prove that a vector perpendicular to all the a_i is perpendicular to every element of V .

Solution. If $v \in V$, write $v = \sum_i c_i a_i$. For a vector w with $w^\top a_i = 0$ for every i , $w^\top v = \sum_i c_i w^\top a_i = 0$. Since v was arbitrary, $w \in V^\perp$. The generators need not be independent; existence of a representation is enough.

Exercise 2.34 (Drawing parallelograms). Draw the parallelograms generated by (a) $(1, 2)$ and $(-1, 1)$; (b) $(2, -1)$ and $(1, 3)$.

Solution. The vertices, in boundary order, are $0, v, v + w, w$. For (a), the vertices are $(0, 0), (1, 2), (0, 3), (-1, 1)$. For (b), they are $(0, 0), (2, -1), (3, 2), (1, 3)$. The full regions, including boundaries, are $sv + tw$ with $0 \leq s, t \leq 1$.



Matrix classification and recurrence problems

Exercise 2.35 (All matrices commuting with a given matrix). For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, show that $AB = BA$

holds exactly for matrices $B = aI + b \begin{pmatrix} 0 & 2 \\ 3 & 3 \end{pmatrix}$, $a, b \in \mathbb{R}$.

Solution. Write $B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. The two products are

$$AB = \begin{pmatrix} p + 2r & q + 2s \\ 3p + 4r & 3q + 4s \end{pmatrix}, \quad BA = \begin{pmatrix} p + 3q & 2p + 4q \\ r + 3s & 2r + 4s \end{pmatrix}.$$

Equality of their entries gives $2r = 3q$, $2s = 2p + 3q$, $s = p + r$ and $3q = 2r$. Hence $r = 3q/2$, $s = p + 3q/2$ with p, q free. Taking $a = p$, $b = q/2$ gives the claimed form. Conversely, I commutes with A , and the second generator is $A - I$ and thus commutes with A . Linearity of multiplication proves the converse for every a, b .

Exercise 2.36 (All square-zero matrices of order two). Show that the only real symmetric 2×2 solution of $A^2 = 0$ is $A = 0$. Describe all real 2×2 solutions as $A = pq^\top$ with $p^\top q = 0$, and show that there are infinitely many.

Solution. If A is symmetric, $A^\top A = A^2 = 0$. Its diagonal entries are sums of squares of its columns, so every entry of A is zero. For general $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the equations are

$$a^2 + bc = 0, \quad b(a + d) = 0, \quad c(a + d) = 0, \quad d^2 + bc = 0.$$

If $b = c = 0$, they give $a = d = 0$. Otherwise $d = -a$ and $a^2 + bc = 0$. If $a = 0$, then $bc = 0$, giving the forms $e_1(0, b)$ or $e_2(c, 0)$. If $a \neq 0$, then $b \neq 0$ and

$$A = \begin{pmatrix} a & b \\ -a^2/b & -a \end{pmatrix} = \begin{pmatrix} 1 \\ -a/b \end{pmatrix} (a \ b).$$

In every case the two outer-product vectors have dot product zero. Conversely $(pq^\top)^2 = p(q^\top p)q^\top = 0$ when $q^\top p = 0$. The family $\begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}$, $t \in \mathbb{R}$, already contains infinitely many distinct solutions.

Exercise 2.37 (Powers preserve symmetry and triangularity). Prove that positive integer powers of a symmetric matrix are symmetric. Prove that powers of a triangular matrix are triangular and that $(A^k)_{ii} = a_{ii}^k$.

Solution. For symmetric A , $(A^k)^\top = (A^\top)^k = A^k$. If A, B are lower triangular and $i < j$, then each summand in $(AB)_{ij} = \sum_h a_{ih}b_{hj}$ vanishes: a nonzero term requires both $h \leq i$ and $h \geq j$, which is impossible. For a diagonal entry only $h = i$ can contribute, so $(AB)_{ii} = a_{ii}b_{ii}$. Induction gives the assertion for A^k . Transposition gives the upper triangular case. The argument applies to every positive integer k , not only to squaring.

Exercise 2.38 (Fibonacci powers and a periodic recurrence). (a) For $F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $x_{-1} =$

$$0, x_0 = 1, x_n = x_{n-1} + x_{n-2}, \text{ prove } F^n = \begin{pmatrix} x_n & x_{n-1} \\ x_{n-1} & x_{n-2} \end{pmatrix} \text{ for } n \geq 1.$$

(b) Put $A = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. Show $B = A^2$, $B^2 = -A$, compute $A^2, \dots, A^6$, and explain how A describes the recurrence $u_n = u_{n-1} - u_{n-2}$.

Solution. (a) For $n = 1$, $x_1 = 1$, giving exactly F . If the formula holds at n , right multiplication by F gives

$$F^{n+1} = \begin{pmatrix} x_n + x_{n-1} & x_n \\ x_{n-1} + x_{n-2} & x_{n-1} \end{pmatrix} = \begin{pmatrix} x_{n+1} & x_n \\ x_n & x_{n-1} \end{pmatrix}.$$

This proves the formula by induction. (b) Direct multiplication gives $A^2 = B$ and $AB = -I$. Hence $A^3 = -I$, $A^4 = -A$, $A^5 = -B$, $A^6 = I$; in particular $B^2 = -A$ and $B^3 = I$. With $z_n = (u_n, u_{n-1})^\top$, the recurrence is exactly $z_n = Az_{n-1}$, since the two coordinates on the right are $u_{n-1} - u_{n-2}$ and u_{n-1} . Thus $z_n = A^n z_0$ and $z_{n+6} = z_n$ wherever the recurrence is defined.

Exercise 2.39 (Idempotent matrices of order two). Prove that the real symmetric 2×2 matrices satisfying $P^2 = P$ are 0 , I , and $aa^\top$ with $a^\top a = 1$. Give a nonsymmetric idempotent $n \times n$ matrix for every $n \geq 2$.

Solution. Write $P = \begin{pmatrix} u & v \\ v & w \end{pmatrix}$. The equations are $u^2 + v^2 = u$, $v(u + w) = v$, $v^2 + w^2 = w$. If $v = 0$, u, w are each 0 or 1 , giving $0, I, e_1 e_1^\top, e_2 e_2^\top$. If $v \neq 0$, $u + w = 1$ and $v^2 = u(1 - u)$, so $0 < u < 1$. Set $a = (\sqrt{u}, v/\sqrt{u})^\top$. Then $aa^\top = P$, because $v^2/u = 1 - u = w$, and $a^\top a = u + w = 1$. Conversely $(aa^\top)^2 = a(a^\top a)a^\top = aa^\top$, and this matrix is symmetric. For the nonsymmetric example take $P = e_1(e_1 + e_2)^\top$ in $\mathbb{R}^{n \times n}$. Since $(e_1 + e_2)^\top e_1 = 1$, $P^2 = P$, but $p_{12} = 1$ and $p_{21} = 0$. No nonsymmetric example exists when $n = 1$.

Exercise 2.40 (Orthogonal matrices of order two). Prove that every real orthogonal 2×2 matrix has one of the forms

$$Q_1 = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad Q_2 = \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix}.$$

Describe their effects on vectors. For general orthogonal Q , prove that its rows and columns each have length 1 and are pairwise perpendicular.

Solution. Write the first row as $(\cos \theta, -\sin \theta)$, possible because its squared length is 1. The perpendicular unit vectors to this row are exactly $(\sin \theta, \cos \theta)$ and its negative: solving the one orthogonality equation gives their line, and the unit-length condition fixes the scalar to 1 or -1 . This gives the two forms, both of which satisfy $Q^\top Q = QQ^\top = I$ by $\cos^2 \theta + \sin^2 \theta = 1$. For a vector of angle ϕ and length r , multiplying by Q_1 gives $r(\cos(\phi + \theta), \sin(\phi + \theta))^\top$, so it rotates by θ . Also $Q_2 = \text{diag}(1, -1)Q_1$, so it rotates by θ then reflects across the horizontal axis. In general, the entries of $QQ^\top = I$ are precisely the row dot products, and those of $Q^\top Q = I$ are precisely the column dot products. Diagonal entries 1 give the lengths, and off-diagonal zeros give perpendicularity.

Exercise 2.41 (Permutation matrices). A permutation matrix has one 1 in each row and each column, all other entries being zero. List all such matrices of orders 2 and 3. Which column permutation changes $A = \begin{pmatrix} 0 & 0 & a \\ 0 & b & 0 \\ c & 0 & 0 \end{pmatrix}$ to $\text{diag}(a, b, c)$? Show there are $n!$ permutation matrices of order n , and prove that each is orthogonal.

Solution. For order 2 the matrices are I_2 and S_{12} . For order 3, specify each matrix by its ordered list of rows; all six are

$$\begin{aligned} (e_1^\top, e_2^\top, e_3^\top), & \quad (e_1^\top, e_3^\top, e_2^\top), & \quad (e_2^\top, e_1^\top, e_3^\top), \\ (e_2^\top, e_3^\top, e_1^\top), & \quad (e_3^\top, e_1^\top, e_2^\top), & \quad (e_3^\top, e_2^\top, e_1^\top). \end{aligned}$$

The last one, S_{13} , gives $AS_{13} = \text{diag}(a, b, c)$. For size n , choose the location of the 1 in the first row in n ways, the second in $n - 1$ remaining columns, and so on. Every choice gives exactly one permutation matrix, so the count is $n!$. Distinct rows have their 1s in distinct columns; their dot products are zero and their squared lengths are 1. Thus $PP^\top = I$, and the same column argument gives $P^\top P = I$.

Exercise 2.42 (Real normal matrices of order two). A real square matrix is normal if $A^\top A = AA^\top$. Show that symmetric and orthogonal matrices are normal. Prove that a normal real 2×2 matrix is either symmetric or has the form $\lambda \begin{pmatrix} \alpha & 1 \\ -1 & \alpha \end{pmatrix}$ with $\lambda \neq 0$.

Solution. For symmetric A , the two products both equal A^2 ; for orthogonal A they both equal I . For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$A^\top A = \begin{pmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{pmatrix}, \quad AA^\top = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix}.$$

The diagonal equations both reduce to $b^2 = c^2$. The off-diagonal equation reduces to $a(b-c) - d(b-c) = (a-d)(b-c) = 0$. If $b = c$, the matrix is symmetric. Otherwise $b = -c \neq 0$, and the second equation forces $a = d$. Take $\lambda = b$, $\alpha = a/b$ to get the stated form. Conversely those entries satisfy both equations, so the displayed family is normal. The cases $b = c = 0$ are already included in the symmetric alternative.

Exercise 2.43 (*A binomial formula without commutation). Let A, B be square matrices of the same order and let p be a positive integer.

- (a) Even when A is invertible, show that $(A + B)^p = A^p(I + A^{-1}B)^p$ need not hold.
- (b) Let $C_{p,j}$ be the sum of all ordered products of p factors containing exactly $p - j$ copies of A and j copies of B . For example, $C_{3,1} = A^2B + ABA + BA^2$. Prove

$$(A + B)^p = \sum_{j=0}^p C_{p,j}.$$

- (c) If $AB = BA$, prove $(A + B)^p = \sum_{j=0}^p \binom{p}{j} A^{p-j} B^j$.
- (d) Show that at $p = 2$ the formula in (c) forces $AB = BA$. Give noncommuting matrices for which that formula holds at some $p > 2$.

Solution. (a) Take $A = \text{diag}(1, 2)$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $p = 2$. Since $A^{-1}B = B$ and $B^2 = 0$,

$$(A + B)^2 = \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}, \quad A^2(I + A^{-1}B)^2 = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}.$$

The factorization $A + B = A(I + A^{-1}B)$ does not permit reordering the factors in its powers.

(b) The claim holds for $p = 1$. Premultiplying the proposed expansion by $A + B$ prepends either A or B to each ordered product. Every word of length $p + 1$ occurs exactly once, according to its first letter. More explicitly, with missing-index terms interpreted as zero,

$$C_{p+1,j} = AC_{p,j} + BC_{p,j-1}.$$

Summing this recurrence proves the next expansion, completing induction. There are $\binom{p}{j}$ words in $C_{p,j}$, because their j positions occupied by B determine them.

(c) Commutation allows adjacent A, B factors to be interchanged until all copies of A precede all copies of B . Every word is then $A^{p-j}B^j$. The count in (b) gives the binomial coefficient.

(d) At $p = 2$, comparing $A^2 + AB + BA + B^2$ with $A^2 + 2AB + B^2$ gives $BA = AB$. For a counterexample at $p = 3$, let $M_{ij} = e_i e_j^\top$ be the 3×3 matrix with its only nonzero entry a 1 in position (i, j) . Take $A = M_{12}$ and $B = M_{23}$. Then $AB = M_{13} \neq 0 = BA$, while $A^2 = B^2 = 0$. The matrix $A + B$ has square M_{13} and cube zero; every term in the proposed cubic binomial sum is also zero. Thus the cubic identity holds although these matrices do not commute.